

Definition of a Function

A **function** is a relation between a set of inputs (domain) and a set of possible outputs (range) where each input is related to exactly one output. Functions can be mathematical or programming constructs, and they are used to model relationships or processes.

Types of Functions

1. Mathematical Functions

- **Linear Function:** $y = mx + c$

Application: Predicting trends (e.g., cost of goods based on quantity).

Example: If $y = 5x + 10$, the cost y for x items increases linearly.

- **Quadratic Function:** $y = ax^2 + bx + c$

Application: Trajectory of projectiles, optimization problems.

Example: The height of a ball thrown is modeled as $h(t) = -5t^2 + 20t + 15$.

- **Exponential Function:** $y = a \cdot b^x$

Application: Population growth, compound interest.

Example: $P(t) = p_0 e^{rt}$, where $P(t)$ is the population at time t .

- **Logarithmic Function:** $y = \log_b(x)$

Application: Measuring pH levels, Richter scale for earthquakes.

- **Trigonometric Functions:** $\sin(x)$, $\cos(x)$, $\tan(x)$

Application: Wave modeling, navigation.

Example: The height of a Ferris wheel seat as it moves can be modeled as $h(t) = R \cdot \sin(\omega t)$

1. Finance:

Exponential Function: Calculate compound interest.

Formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

Question: If $P = 1000$, $r = 5\%$, compounded annually ($n = 1$) for 3 years ($t = 3$), what is A ?

Solution:

$$A = 1000 \left(1 + \frac{0.05}{1} \right)^3 = 1000 \cdot 1.157625 = 1157.63$$

The total amount after 3 years is \$1157.63.

2. Physics:

Quadratic Function: Modeling a projectile's motion.

Formula:

$$h(t) = -4.9t^2 + v_0t + h_0$$

Question: If a ball is thrown upward with an initial velocity of 20 m/s from a height of 5 m, when will it hit the ground?

Solution:

Solve $-4.9t^2 + 20t + 5 = 0$.

Using the quadratic formula,

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-20 \pm \sqrt{400 + 98}}{9.8} = 4.52 \text{ seconds.}$$

The ball hits the ground after 4.52 seconds.

3. E-commerce:

Linear Function: Predicting shipping cost.

Scenario: Shipping cost y is calculated as $y = 10x + 5$, where x is the number of items.

Question: What is the cost to ship 3 items?

Solution:

$$y = 10(3) + 5 = 35$$

The shipping cost is \$35.

1. Injective Function (One-to-One)

A function $f : A \rightarrow B$ is injective if every element of the domain A maps to a unique element in the range B . In other words, no two different inputs in A produce the same output in B .

Mathematical Definition:

For $f(x_1) = f(x_2)$, it must follow that $x_1 = x_2$.

Example:

$f(x) = 2x$ is injective because each x produces a unique output.

Application:

- **Cryptography:** Injective functions are used to ensure that encryption maps each input (plaintext) to a unique output (ciphertext).
- **Database Systems:** To avoid redundancy, unique identifiers (like Social Security Numbers) are mapped injectively to individuals.



2. Surjective Function (Onto)

A function $f : A \rightarrow B$ is surjective if every element in the codomain B is the image of at least one element in the domain A . In other words, the function "covers" the entire range.

Mathematical Definition:

For every $y \in B$, there exists at least one $x \in A$ such that $f(x) = y$.

Example:

$f(x) = x^2$ over the domain $x \in [-\infty, \infty]$ is not surjective for $f : \mathbb{R} \rightarrow \mathbb{R}$, because negative values are not in the range. However, $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ is surjective.

Application:

- **Resource Allocation:** Assigning every available resource to at least one task.
- **Network Design:** Ensuring every node in a network receives data (e.g., broadcasting).

3. Bijective Function (One-to-One and Onto)

A function $f : A \rightarrow B$ is bijective if it is both injective (one-to-one) and surjective (onto). In this case, there is a perfect pairing between elements in the domain A and codomain B .

Mathematical Definition:

For every $y \in B$, there exists a unique $x \in A$ such that $f(x) = y$.

Example:

$f(x) = x + 3$ for $x \in \mathbb{R}$ is bijective because every real number is mapped uniquely and every real number is covered.

Application:

- **Reversible Processes:** Temperature conversion (e.g., Celsius to Fahrenheit).
- **E-commerce:** Mapping a unique order ID to each customer purchase.

Applications in Real-world Contexts

Injective Function Application:

- **Data Security:** Encryption ensures each key encrypts data uniquely.
- **Database Management:** Assigning unique usernames to users.

Surjective Function Application:

- **Resource Allocation:** Ensuring all tasks are assigned workers (not necessarily uniquely).
- **Mathematical Modeling:** For any input, ensuring that all possible outcomes (range) are covered.

Bijective Function Application:

- **Cryptographic Decoding:** Each ciphertext corresponds to a unique plaintext, making decryption possible.
- **Indexing Systems:** Mapping library books to unique shelf locations.

Example Problems

Problem 1: Injective Function

Prove that $f(x) = 3x + 2$ is injective for $x \in \mathbb{R}$.

Solution: Assume $f(x_1) = f(x_2)$:

$$3x_1 + 2 = 3x_2 + 2$$

Subtract 2:

$$3x_1 = 3x_2$$

Divide by 3:

$$x_1 = x_2$$

Hence, $f(x)$ is injective.

Problem 2: Surjective Function

Show that $f(x) = x^2$ for $x \in \mathbb{R}$ is not surjective if the codomain is $\mathbb{R}$.

Solution: The range of $f(x) = x^2$ is $\mathbb{R}_{\geq 0}$, as the square of any real number is non-negative. Since negative numbers in $\mathbb{R}$ are not included, $f(x)$ is not surjective.

Problem 3: Bijective Function

Prove that $f(x) = 2x + 1$ is bijective for $x \in \mathbb{R}$.

Solution:

- **Injective:** Assume $f(x_1) = f(x_2)$:

$$2x_1 + 1 = 2x_2 + 1 \implies x_1 = x_2$$

Hence, $f(x)$ is injective.

- **Surjective:** For any $y \in \mathbb{R}$, solve $y = 2x + 1$ for x :

$$x = \frac{y - 1}{2}$$

Since $x \in \mathbb{R}$ exists for all y , $f(x)$ is surjective.

Thus, $f(x)$ is bijective.